FIN 433

FIN 412/433: Theory of Corporate Finance

1. Capital market, Consumption and Investment decision.
2. Capital Investment Analysis Revisited. *NPV vs IRR *Different live problem * Treatment of inflation
3. Capital Investment Analysis with Uncertainty Revisited. * Dependent proje
4. Capital Investment with Imperfect market. * Capital Rationing and LP
5. Capital Investment: * taxation and financing * treatment of tax * Lease or buy decision.
6. Utility theory.
7. Mean Variance theorem.
8. Capital market efficiency.
9. Capital structure decision: further topic.
10. Dividend decision: further topic.
11. Interaction of financing and Investment decision.
12. Introduction to options.

Recommended Textbook: i) Lowe text book. (Pdf)

ii) Financial theory and corporate policy.

MRS ⇒ is the rate at which you are willing to reduce the consumption of one goods to get one more unit of another good and still remain indifferent.

At equilibrium the slope of indifference curve = Slope of the budget line.

Global Topic: Capital Market, consumption and Investment Decision.

What is a capital market: This is a market for long term loans & long term funds.

Assumptions: Underlying of a perfect market

- Information is readily and freely available.

2. free entry and exist etc 3. free flow of information

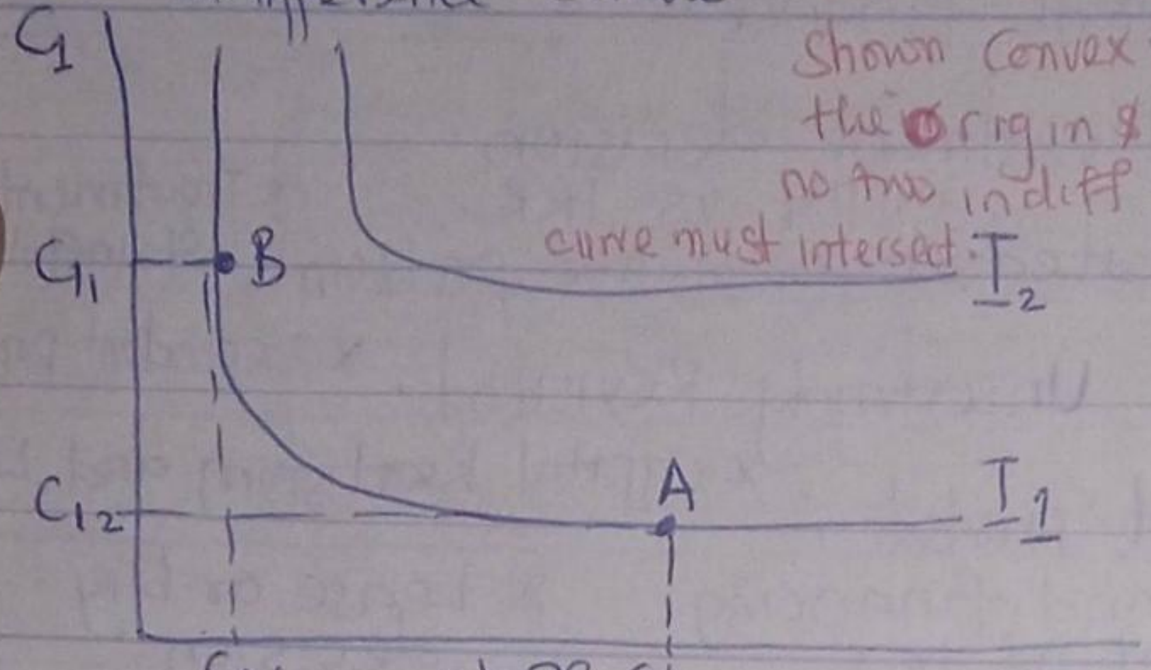
4. No tax, 5. No transaction cost

Illustration: for two commodities Y_0 & Y_1,

- Indifference curve shows the difference btw the preference today & preference tomorrow

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An indifference curve.



To find the slope of the Indifference curve, draw a target to the curve. Measurse the Marginal rate of substitution (MRS)

Note: The higher level of Indifference curve, the higher the satisfaction. (Because more goods are consumed.)

Marginal rate of substitution will be derived when a line is drawn.

Note: The indifference curve comprises of consumption today and consumption tomorrow.

$$MRS = \frac{\partial C_1}{\partial C_0} = \text{f(...)} \div (1+r).$$

production possibility or Investment opportunity available in the economy.

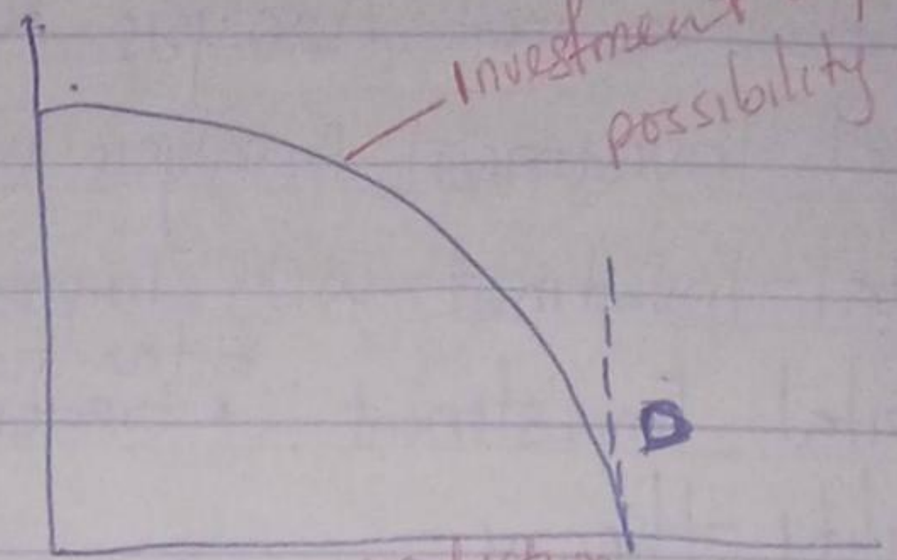
The more the Investment the more the returns as in any economy Individuals Invest in other to get returns.

Note: The slope of an Indifferencce curve is known as Marginal rate of substitution.

* The marginal rate of substitution is also known as "Subjective rate of time preference"

Note: Indifference curve are shown convex to the origin & no two indiff curve must intersect.

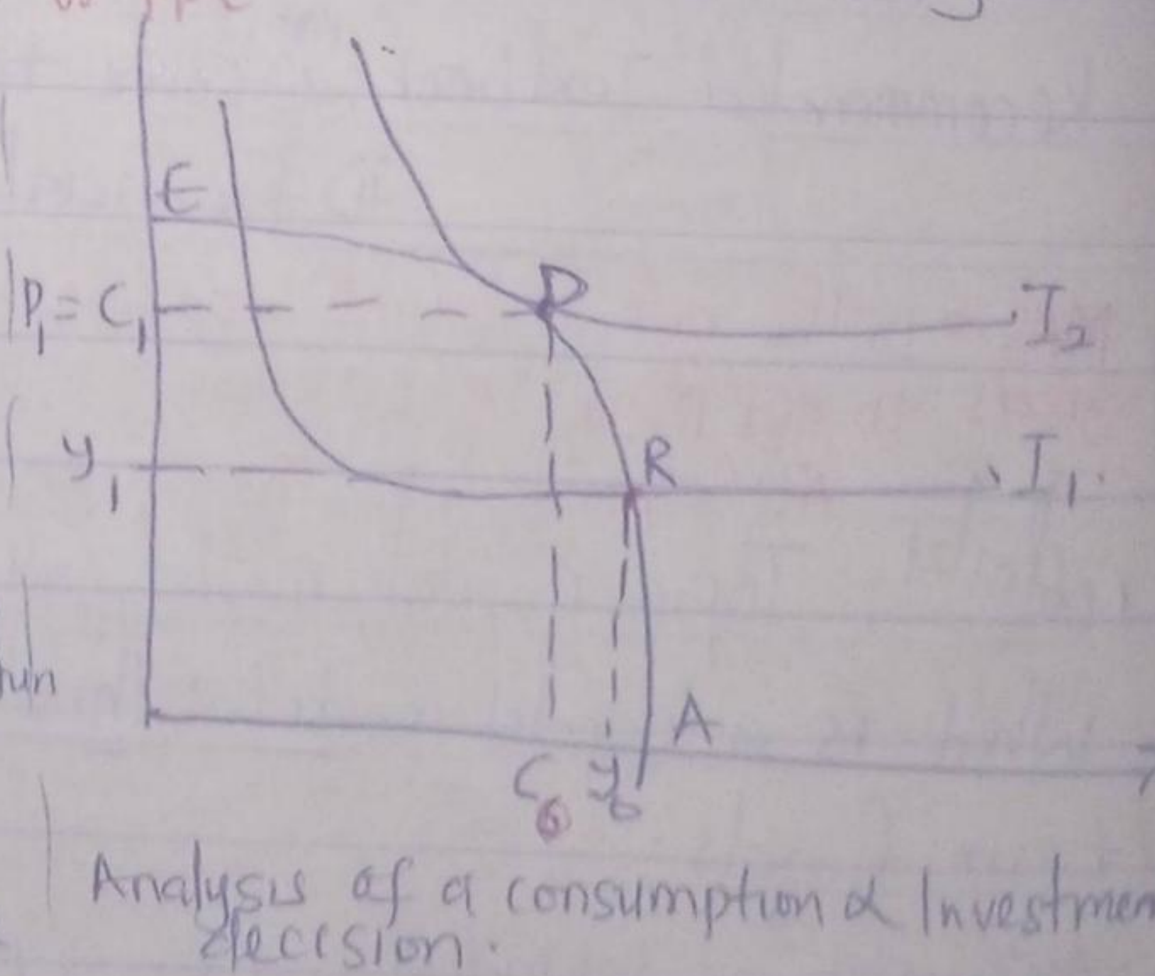
* The law of diminishing marginal level of Investment



Investment or production opportunity curve.

The slope of the P.P.C or I.P.C is the Marginal rate of transformation; The MRT is the number of unit of goods that must be forgone in order to create one unit of another goods. (two)

Combination of the Investment Indifference Curves and Investment opportunity curve or PPC.



Analysis of a consumption & Investment decision.

At point "D" is the "optimal consumption decision point"

Note: at point "D" satisfaction is at maximum.

Note: o

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The difference btw C_0 and Y_0 is Investment since the Investor did not finish his wealth as he has moved to a higher level of Investment and he is bound to consume higher tomorrow.

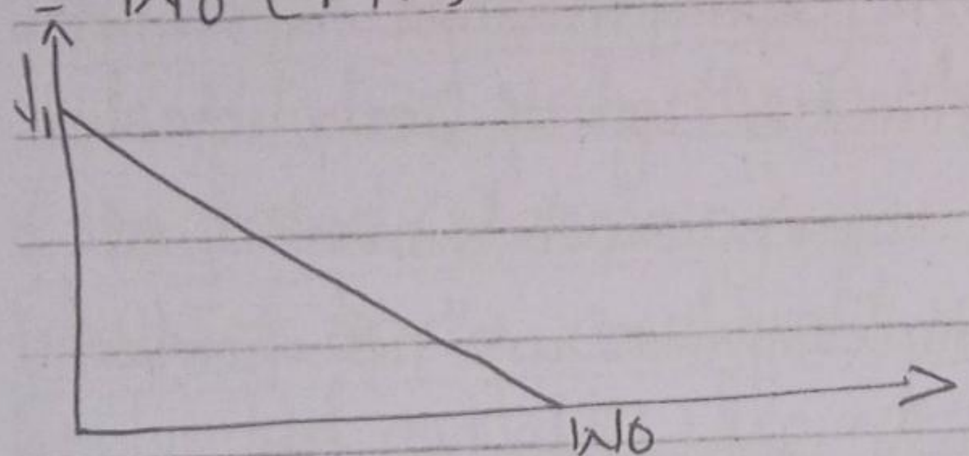
∴ At point D; MRT = MRI.

Marginal rate of Transformation = Marginal rate of Consumption.

In capital market there's possibility of C.M exchange, It will facilitate the transfer of resources (funds) from the surplus to the deficit units.

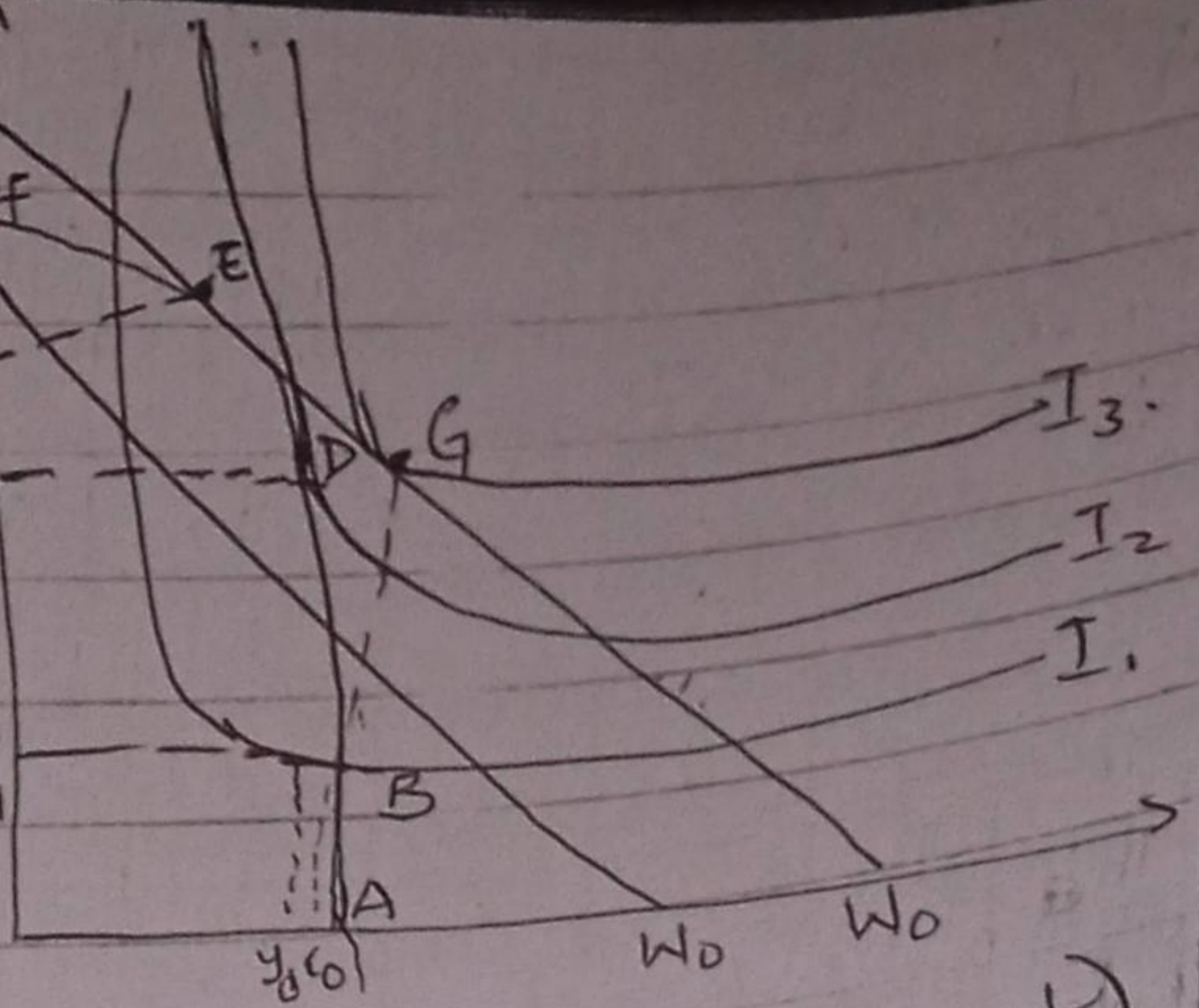
Note: With capital market we assume a perfect capital market and we are to assume that the borrowing rate and lending rates are the same.

Graph illustrating the capital market $= W_0(1+r)$.



The slope of the line is "the market rate of interest".

Note: We want to combine the capital market and the previous graphs drawn earlier.



At point B, (Y_0, Y_1 is at this point), has three choices i.e the investment point (A, B, D, E, f) → Investment opportunity curve.

Point (W_1, W_0) → Capital market curve

Point (I_1, I_2, I_3) Indifference curve.

Explanation of the curve.

The Investor has choice to Invest in any of the three but at point "D" the Investor can borrow money from C.M and continue the Investment until he reaches point E which is the optimal point The slope of Indifference curve is tangent to the slope of the capital market curve.

At point E, Investor can lend or borrow to the C.M until he reaches the Point E.

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At point G, the slope of C1M line is target to the slope of Indifference curve three (I_3) or Indifference curve I_3 is target to the capital market curve

Note! Point G is the Optimal Consumption decision. It is the point where MRT = MRI (Marginal rate of Transformation = Marginal rate of Investment

This is the theory is the fishers seperation theory:

* Given perfect and complete capital market, production decision is govern solely on the basis of subjective market criteria without reference to the Individual subjective rate of time preference which enters into his consumption decision.

Note: With capital market, Individual consumes more and will move to a higher Indifference curve. The capital market (funds) enables efficient allocation of resources. from the surplus sector to the deficit sector.

Note: from the graph above, the point "E" was not considered as the optimal Consumption decision point because it is not target to any Indifference curve.

Indifference Curve shows a combination of two commodities that gives a consumers equal satisfaction and utility thereby making the consumer indifferent. It makes the consumer practice "trade off"

Topic: Capital Rationing (Multi-Stage).

capital rationing can be defined as the situation in which companies or firm are unable to accept projects with positive NPV due to limited financial resources.

Causes of Capital Rationing:

These can be categorised into

1) External causes;

a) Government restriction on bank lending through CBN. (i.e the banks are not supportive in lending funds for

b) Imperfection in Nigeria capital market (due to this people cannot raise enough capital).

c) High floatation cost charged in Nigeria capital market. (floatation cost is the cost of been listed in the Nigeria capital market as many companies are not quoted.

d) High administration charges in Nigeria capital market.

2) Internal causes:

Note: Once there is change in a companies Mgt there is bound to be capital rationing.

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changes in companies management
changes in technology
Reduction in compabaies sales
Drop or reduction in company profit

Years	Projects: A ₦'000	B ₦'000	C ₦'000	D ₦'000	E ₦'000
2022	(8000)	(10,000)	(20,000)	(4000)	(15,000)
2023	(3000)	(4000)	(2,000)	(1000)	(6000)
2024	2,000	6,000	10,000	800	7000
2025	3,000	7,000	8,000	1400	8000
2026	5,000	8,000	8,600	1600	9000
2027	10,000	3,000	6,000	2000	9600

Types of project in Caprtal Rationing

Indivisible project: That is, a Complete whole project must be undertaken

Divisible project: That is a fractional part of a project can be undertaken

Steps Involve in multi-Stage Capital rationing

Step 1: Determine the NPV of all projects under consideration.

Step 2: Cancel out project(s) with negative NPV.

Step 3: Then set up the linear programming model by formulating the objective function using the positive NPV of the project(s) remainging

Step 4: Set up the constraints using the limited available resources.

Step 5: set up the non-negativity constraints.

Illustration 1: Rebecca Limited have estimated the expected Investment outlay and return for 5 (five) possible projects as follows:

The company cost of capital is 15% and the projects are divisible. The company has a maximum of 20 million to Invest now. (2022) and a further 5 million in one year time (2023). Required:

a) formulate, but dont solve, the linear programming problem to select the optimal combination of project [exam question: 6, 8, 9, and 10].

Solution.

Project A.	Year	Discounted cash flow D.C.f %
	0	
	1	
	2	
	3	
	4	
	5	

Using: $\frac{1}{(1.15)^1}$

P.T.O

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Year	project A	DCF 15%	PV A	project B	PV B	Project C	PV ₦20,000 C	project D ₦4,000	Present value PV D	project E	Present value
0	(8000)	1.0000	(8000)	(10,000)	(10,000)	(20,000)	(20,000)	(4000)	(4000)	(15,000)	(15,000)
1	(3000)	0.8696	(2609)	(4000)	(3,478)	(2000)	(1739)	(1000)	(870)	(6000)	(5218)
2	2000	0.7561	1512	6000	4,537	10,000	7561	800	605	7000	
3	3000	0.6575	1973	7000	4,603	8000	5260	1400	921	8000	5260
4	5,000	0.5718	2859	8000	4,574	8600	4917	1600	915	9000	5146
5	10,000	0.4972	4972	3000	1492	6,000	2983	2000	994	9600	4,773
			707,000		1,728,000		(1018)		(1,435,000)		₦254,000

Note: Cancel project C since its NPV is Negative and also project D should be cancel

Note: The D.C.F for all the projects remain the same. To obtain the DCF use this formular $\frac{1}{(1+r)^n}$ or $\frac{1}{(1+0.15)^n}$ for project A under year (0) but anything raise to power (0) is 1 $\therefore \frac{1}{(1.15)^0} = 1,000$

MAX Z $\longrightarrow$ ₦707,000 A + ₦1,728,000 B + ₦254,000 E

Subject to (s.t): 2022 Constraint: $\rightarrow$ 8,000,000 A + 10,000,000 B + 15,000,000 E $\leq$ 20,000

Subject to: 2023 Constraint: $\rightarrow$ 3,000,000 A + 4,000,000 B + 6,000,000 E $\leq$ 5,000

Note: A, B, E $\geq$ 0 i.e the three projects must not be negative.

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Prof T.M. Obamuyi

27/01/22.

pic: Capital Budgeting

apital budgeting refers to the organisation planning process to invest its vest funds most efficiently in long term asset in anticipation of an pected flow of benefits over serials of years.

Capital budgeting enables us to evaluate projects to know if it is desirable or not.

e: NPV and IRR are the most important techniques under C. budgeting.

re are possibilities for both NPV and IRR to give you the same decision.

$NPV = \sum_{t=0}^{n} \frac{C_{f_t}}{(1+k)^t}$ at year 0.; while $NPV = \sum_{t=1}^{n} \frac{C_{f_t}}{(1+k)^t} - I_0$ < year 1

< Initial year

IPV ≥ 0 or + (positive) < Accept).

IPV < 0 or − (negative) (Reject).

Note I_0 → Initial outlay.

$G = NPV = \sum_{t=1}^{n} \frac{C_{f_t}}{(1+r)^t} - I_0$ < i.e what % of r will make the NPV "0".

Comparison of NPV and IRR.

Both NPV and IRR will give the same result < i.e acceptance & rejections) garding an investment proposal in the following situations

When the project under consideration involves conventional cash flow e only one change in the cash flow sign. The (−) = Outflow, while (+) inflow.

When the projects are independent of one another i.e the acceptance one does not preclude the acceptance of the other and if the firm is i facing a problem of fund constraint. eg When we have 3 projects with ch we NPV i.e greater than Zero) then the three projects should be taken.

Causes of NPV and IRR Conflict.

ion Conventional cash flow or multiple rates of return. eg if the mpany has (+/−/+/+) and other project < − / + / − / +) The project can agree.

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Eg. C_0 C_1 C_2 NPV 25% ; NPV 400%
(800,000) 5,000,000 (5,000,000) ≠0 ≠0

Note: With is there will be possibility for the company to have two rates of return, the question is which rat of return will you choose.

In the above table, NPV is "0" at two discount rate which is at 25% and 400%.

A cash flow stream with "M" changes in sign, you can have upto M positive (+M) Internal rate of returns. In other words, the number of IRR depends on the number of times the sign of the cashflow changes eg − + ÷ (equal 2 IRR). for this situation. It is better to use NPV method to evaluate the project given above.

2. Lending vs borrowing project:

	C_0	C_1	NPV	IRR	
Project P =	(20,000)	30,000	7,272.73	50%	(NPV determi
G =	20,000	(35,000)	(11,818.18)	75%	

for project P, we are Investing (20,000) immediately before the cashflow comes in for project G; We are not investing immediately, we have to borrow the ~~already have inflow of~~ ₦20,000

Note: project "G" can either borrow for govt, firm or even give out the project to any one that has money.

Note; Project P is "Investing" while project G is borrowing

Note: R → rate prevaling in the economy. (discount rate

IRR → rate of return on Investment.

* IRR > r Accept a project

Note: Accept the project When IRR is less than discount rate (IRR < K)

Just for the Reject the project when IRR is greater than discont rate (IRR > K)

Note the above conclusion negates the standard or decision of IRR.

Note: This Assumption is used only when we have lending & borrowing projects.

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Eg. C_0 C_1 + C_2 — NPV 25% + NPV 400%
(80,000) 500,000 (500,000) Note: with is there will be possibility for the Company to have two rates of return, the question is which of rate will you choose.

In the above table, NPV is "0" at two discount rate which is at 25% and 400%.

A cash flow stream with "M" changes in sign, you can have upto M positive (+M) internal rate of returns. In other words, the number of IRR depends on the number of times the sign of the cashflow change eg − + ÷ (equal 2 IRR). for this situation. It is better to use NPV method to evaluate the project given above.

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Just for this question
Reject the project when IRR is greater than discont rate (IRR > K)

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Note: for the lending and borrowing projecting, we go with the NPV because it is better than IRR.

For Mutually exclusive project; These are projects that the acceptance of one precludes the acceptances of the others. There are three senerio which are:

1.) Where the projects requires different outlay (investment
2.) Where the projects have different unequal lifes.
Where the projects have different patterns of cash flow.

Illustration:

a) Different cash outlay.

	C_0	C_1	C_2	C_3	NPV (10%).	IRR
X	(16,000)	12,000	7,000	5,000	4,450·79	28%
Y	(160,000)	40,000	70,000	120,000	24,372·65	17%.

Suppose we have these projects with conflicting interest based on the NPV and IRR; we will be using "Incremental Approach" i.e (Y-X).

	C_0	C_1	C_2	C_3	NP	IRR
Y-X =	(144,000)	28,000	63,000	115,000	19,921·86	16%.

Accept project Y → because it gives a positive NPV ₦19,921·86 and an IRR of 16% that is greater than the cost of capital of 10%. & Note that project Y includes all the benefit in project X.

b) Unequal lives of project:

	C_0	C_1	C_2	C_3	C_4	NPV 10%	IRR
A	(25,000	30,000	0	0	0	2,273	20%
B	(25,000)	0	0	0	43,750	4,882	15%.

To solve this use "Incremental Approach." Note: for project, it is so difficult to determine the largest project.

c) Different patterns of cash flow: Under this we have different cash flows.

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	C_0	C_1	C_2	C_3	NPV 10%	IRR
X	(16,000)	12,000	7000	2000	2,196.84	20%
Y	(16,000)	4,000	8000	12,000	3,263.71	19%

Note: The bigger project is "Y" irrespective of the initial outlay using Incremental Approach. (Y-X).

	C_0	C_1	C_2	C_3	NPV 10%	IRR
Y-X =	0	(8,000)	1000	10,000	1,066.87	18%

Decision: Accept project Y since the NPV is positive and the IRR is greater than the cost of capital. The project Y still includes all the benefits in project X

Note: Wherever conflicts arises btw NPV and IRR always accept the project with higher NPV.

*The net present value is more realizable than internal rate of return for ranking two or more mutually exclusive capital budgeting problem.

*The result suggested by NPV method is more reliable bcos of the objective of the company to maximise it's shareholders wealth.

*NPV considered the total yield on investment, IRR assumes that any cashflow can be re-invested at the internal rate of return. This assumption is problematic and the NPV does not suffer from such problematic assumption.

osbolaa

construction of last class.

Year	Cashflow	Discounting factors ($r = 25\%$)	Present value
0	-80,000	1	-80,000
1	500,000	0.8000	400,000
2	-500,000	0.64,000	-320,000
		NPV	0

Note: the rate makes NPV = 0 is the IRR = 25%

{Discounting factor = $\frac{1}{(1+r)^n}$ (Lumpsum) or $(1+r)^{-n}$}

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Illustration	C_0	C_1	C_2	C_3	NPV 10%	IRR
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2	-500,000	0.64,000	-320,000
		NPV →	0

Note: the rate makes NPV = 0 is the IRR = 25%.

Continuation of last class.

{Discounting fact = $\frac{1}{(1+r)^n}$ (Lumpsum) or $(1+r)^{-n}$}

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Compare the N.P.V (When we have Project with unequal life to do:
Convert the projects to annuity: Cashflow
replicate the projects until they have the
same life.

se1

	C_1	C_2	C_3	C_4
(20,000)	50,000		0	1000
(20,000)	70,000	100	0	0

ase2.

0	1	2	3	4	5	6
(70,000)	28,000	33,000	38,000	—	—	—
(85,000)	35,000	30,000	25,000	20,000	15,000	10,000

alculate the NPV. (Project C)

Cf	$Df_{10\%}$	PV.
(70,000)	1.00000	(70,000)
28,000	0.9091	25,454.8
33,000	0.8264	27271.2
38,000	0.7513	28,549.4
	NPV:	11,278.40

C.f	Df 10%	PV Project(D)
(85,000)	1.00000	(85,000)
35,000	0.9091	31,818.5
30,000	0.8264	24,792
25,000	0.7513	18782.5
20,000	0.6830	13,660
15,000	0.6209	9313.5
10,000	0.5645	5645
	NPV →	19.011.5

NPV C: 11,278.40 IRR% 20%
NPV D: 19,011.50 15%.

Normally, the best project is project D with the highest NPV although the two projects, carry be carried out. but the best method to use are the following.

Replication method

1) Replicate the project i.e starting the project again in year 3. for project C.

C	0	1	2	3
	(70,000)	28,000	33,000	38,000

Replicate the project in (70,000) third year. = 70,000 + 38,000 = 32,000

C=	0	1	2	3	4
	(70,000)	28,000	33,000	-32,000	28,000
	5	6			
	33,000	38,000			

Step 1: calculate a new NPV for this project.

Year	C.f	Df 10%	PV
0	(70,000)	1.0000	(70,000)
1	28,000	0.9091	
2	33,000	0.8264	
3	(32,000)	0.7513	
4	28,000	0.6830	
5	33,000	0.6209	
6	38,000	0.5645	
		NPV →	

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After replication, the new NPV was ₦19,749.10 for project C while the project D NPV remains ₦19,011.50.

Note: The best project to undertake now is project C since the two projects now have equal life of 6 years and project C has the highest NPV.

Using Annuity method.

$$\text{Annual NPV}_C = \frac{\text{NPV}}{\text{Annuity discounting factor}}$$

$\frac{[1-(1+r)^{-n}]}{r}$ $r = 10\%$ $n = 3$

$$\frac{11,278.40}{2.4869} = ₦4535.12$$

$$\text{Annual NPV}_D = \frac{19,011.50}{4.3553} = 4365.14$$

$r = 10\%$; $n = 6$

Note: project C with the higher NPV using the two approaches is the best project to embark on, therefore do not consider the first calculation where the two projects have different life span, because this first approach make project D have higher NPV.

solving for Illustration 3 of the question

Yr	C.f	Df 10%	PV
0	0	1.0000	0
1	(8000)	0.9091	(7272.8)
2	1000	0.8264	826.4
3	10,000	0.7513	7513
		NP	1066.6

That is how, the NPV was obtained.

PV
0
(6666.4)
694.4
5787
NPV (-185)

Note: The two rates to use in other to calculate your IRR is 10% and 20%

Using Interpolation Approach.

$$IRR = L_r + \left[\frac{NPV_P}{NPV_P + NPV_N}\right](H_R - L_R)$$

$$0.10 + \left[\frac{1066.6}{1066.6 + 185}\right] \times [0.20 - 0.10]$$

$= 18.5\%$

Note: A positive NPV is also equivalent to an "Unrealised Capital gain"

Note: the IRR is the rate that will make the project to give an NPV of zero.

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Abass Shiro 10/02/22

pic: Capital Budgeting with Inflation:

1 is inflation: Inflation can be defined the continous or persistence increase general price level (prices of goods & services) in an economy.

Note: 1) The definition above can be ided as general inflation

In corporate finance inflation can divided into "two (2) concept" namely General inflation and Specific Inflation.

e: The general inflation is affecting whole economy as defined above.

Specific Inflation is inflation that affecting a project under considera on that is, Increase in prices of element of the projects are cashflow Naira and cost in Naira. Specific lation is the most important inflation der corporate finance.

e cash inflows and the cost of the ject under consideration. In specific inflation only affecting cash inflows and the cost that is, both cash inflows and cash outflows).

5) General inflation affecting cashinflows, cost and the cost of capital.

6) There are two types of cash inflows:

(a) Money cashflows (MCf) = Nominal cash flow or cash flows without inflation.

(b) Real cash flow (RCf): Thus is defined as Money cashflow / (1 + Inflation Rate)^T

where T -> Time. [It does not have inflation

7). There are two types of cost of capital.

(a) Money cost of capital (MCC)

MCC = K where K is cost of capital.

b) Real cost of capital (RCC)

$$RCC = \left(\frac{1+k}{1+\text{inflationrate}}\right) - 1$$

where K = Cost of capital.

8:) Money cashflows must be discounted by money cost of capital

9) Real cash flows must be discounted at Real cost of capital.

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10*) The effect of specific inflation shall always be reflected in the cash flow. The cash flow obtained (arrived at) should be discount at the relevant cost of capital which in most cases is the money cost of capital.

Illustration 1: (Chapt (4)): The expected annual rate of inflation is 10% and the money cost of capital is 20%. Calculate the real cost of capital.

solution

$$RCC = \left\{ \frac{(1+k)}{[1+\text{inflation rate}]} \right\} - 1$$

$$\frac{1.20}{1.10} - 1$$

$$= 0.09 \approx 9\%$$

Illustration 2: Kalu Nig limited is considering a project with initial cost of ₦23,000, the life of the project is expected to be 4 years. while the expected scrap value is Nil (0). The annual cash inflows including the effect of inflation are as stated below. solve question

Year	1	2	3	4
Cashflows	15,000	21,000	23,000	17,000

The market discount rate is 18% per annum and all cash flows affected by constant inflation 5% per year (annum).

Required: (a) calculate the NPV of the project based on.

i) Money cash flows
ii) Real cash flows.

b) calculate the real internal rate of return (IRR)

solve question 3, 4, 5 chapter 4

solution:

Year	Cashflow	D.C.F 18%	Present value
0	(23,000)	1.000	(23,000)
1	15,000	0.8475	12,713
2	21,000	0.7182	15,082
3	23,000	0.6086	13,998
4	17,000	0.5158	8,769
			27,562

Note: Under After solving using the MCF, you have to discount to get the Real cash flow.

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10*) The effect of specific inflation shall always be reflected in the cash flow. The cash flow obtained/arrived at) should be discount at the relevant cost of capital which in most cases is the Money cost of capital.

Illustration 1: (Chapt (4)): The expected annual rate of inflation is 10% and the money cost of capital is 20%. Calculate the real cost of capital

solution

$$RCC = \left[\frac{(1+K)}{[1+\text{inflation rate}]}\right] - 1$$

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solution

$$RCC = \left[\frac{(1+K)}{1+\text{inflation rate}}\right] - 1$$

$$\frac{1.20}{1.10} - 1$$

$$= 0.09 \text{ or } 9\%$$

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Year	1	2	3	4
Cash flows	15,000	21,000	23,000	17,000

The market discount rate is 18% per annum and all cash flows are affected by constant inflation at 5% per year (annum).

Required: (a) calculate the NPV of the project based on.

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solve question 3, 4, 5 chapter 4

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Note: Under After solving using the MCf, you have to discount to get the Real cash flow.

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cashflow.

$= \frac{15,000}{1.05} = 14,286.$

$2 = \frac{21,000}{(1.05)^2} = 19,048$

$3 = \frac{23,000}{(1.05)^3} = 19,868$

$4 = \frac{17,000}{(1.05)^4} = 13,986.$

Using R = 62%

Year	Cash flow	D.Cf 62%	PV
0	(23,000)	1.0000	(23,000)
1	14,286	0.6173	8819
2	19,048	0.3810	7257
3	19,868	0.2352	4673
4	13,986	0.1452	2031
			(220)

using D.Cf 52%.

Year	Cash flow	D.Cf. 52%	
0	(23,000	1.000	(23,000)
1	14,286	0.6579	9399
2	19,048	0.4328	8244
3	19,868	0.2848	5658
4	13,986	0.1873	2620
			2921

Cash flow capital (RCC).

$RCC = \left[\frac{1+K}{1+\text{Inf rate}}\right] - 1$

$RCC = \left[\frac{1.18}{1.05}\right] - 1 = 0.12$

$= 12\%.$

lot a new table with the Real flow and Real cost of capital.

	cash flow	D.C.f 12%	present vale
	(23,000)	1.0000	(23,000)
	14,286	0.8929	12,756
	19,048	0.7972	15,185
	19,868	0.7118	14,142
	13,986	0.6355	8,888
			27,971

Calculate the IRR,.

$RCC = LR + \left(\frac{NPV_+}{NPV_+ + NPV_-}\right) \times (H_R - L_R)$

$RCC = 0.52 + \left(\frac{2921}{2921+220}\right)(0.62 - 0.52)$

$0.52 + \frac{2921}{3141}(0.1)$

$= 0.61$

Note: when doing IRR, compare 2 cost of capital that will give you NPV close to zero but not far from each other

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use 62% to get −NPV and 52% to get +NPV. Advance way to get IRR, Pick a cost of capital where you get the −NPV then pick a COC less than the previous COC by 10% to get +NPV. eg 62% → −NPV
52% → +NPV

Illustration 3: Agbo limited is considering a project cost ₦600,000. The project is expected to have a life of 4yrs with a scrap value of ₦48,000. Annual cash revenue from the project is expected to be ₦400,000 in year 1 raising by 6% per annum for inflation. Running cost is expected to be ₦180,000 in year 1 raising by 14% per annum because of inflating labour cost. The general rate of inflation is expected to be 7% and that company money cost of capital is 20%
Required: Advise the company on the acceptability of the project.

Solution:

Exam: q4 (1,2,3,4 & 5) Exam question

Step: Calculate the Revenue first.

Year 1: Revenue = ₦400,000 for all

(Note: If the question is say the revenue is ₦400,000 for 2% then year 1 = 400,000(1.06), year 2 = 400,000(1.06)²)

year 2: ₦400,000 (1.06) = 424,000
year 3: 400,000 (1.06)² = 449,440
year 4: ₦400,000 (1.06)³ = 476,406

Running cost.

year 1: ₦180,000 → 180,000
year 2: 180,000(1.14) → 205,200
year 3: 180,000(1.14)² → 233,928
year 4: 180,000 (1.14)³ → 266,678

Note: put cost in year 0 and put scrap value in year (4) the last year. put the Running cost in bracket (It is cash outflow).

year	cost & scrap val	Revenue	Running cost	Net cash flow (NCF)	DCF 20%
0	(600,000)			(600,000)	1.0000
1		400,000	(180,000)	220,000	0.8333
2		424,000	(205,200)	218,800	0.6944
3		449,440	(233,928)	215,512	0.5787
4	48,000	476,406	(266,678)	257,728	0.4823

Note: you to discount based on the money cost of capital 20% i.e from Note 10 above.

year	0	1	2	3	4
PV:	(600,000)	183,326	151,935	124,717	124,302

NPV (15,720)

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mment: Since the NPV of the proposed
roject is Negative, the company should
ot accept the project.

ote: Money Cash flow has Inflation will
eal cash flow does not
~~Assignment~~ have Inflation
g The price of calculators last years
as ₦2000, but the present price now is
₦4000 due to inflation, the M'CC is ₦4000

Assignment.

Ekpoma Ventures Nigeria Limited is
onsidering a project costing ₦60,000.
e project is expected to have a life
pan of 5 yrs. with residual
alue of ₦50,000. The selling price
er unit of the product will be ₦12
n year 1 and raise by 10% per
nnum as a result of inflation. The
oject also involves labour cost of
₦6 per unit in year one (1) and
aise by 9% as a result of inflation
Thes fixed operating cost is expec
ted to be ₦100,000 in year 1
and raise by 8% per annum
as a result of inflation. The
general price level increase (genera
inflation) in the country is 12%
and the company cost of capital
is 20%. Required advise
the company management
whether or not to accept the
project using the NPV cottena.
The expected annual sales
demand is 35,000 unit.

26/10/22. Prof Olowe.

Topic:
Capital Market consumption
and Investment Decision.
capital market is a market for
medium and long term funds
are raised. It can be security
based or non-security based.
Capital To under this C.M. Cwere
are going to be using an assum
-ption of perfect market.
In a perfect market;
* Informationare freely available
* free entry and free exist.
* No taxes * No transaction cost
*
*
We need information to enable us make
sacrifice & we need information on
production pussibility availalble in
an economy

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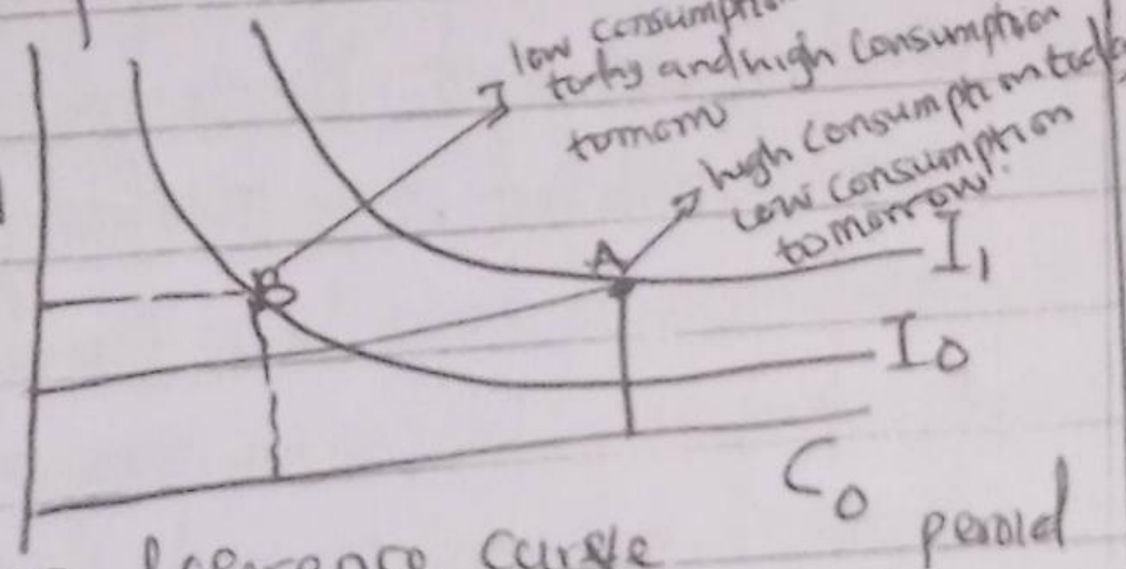
you can either consume C_0 today or C_1 tomorrow. There will be Comparism btw C_0 and C_1.

* We are assuming one man economy

Income : Y_0 and Y_1

* Assume that marginal utility of consumption is increasing at a decreasing rate.

* Utility of of consumption curve for period 1 Constant. (all these are assumptions)



Indifference curve

This shows the preference btw today and tomorrow consumption.

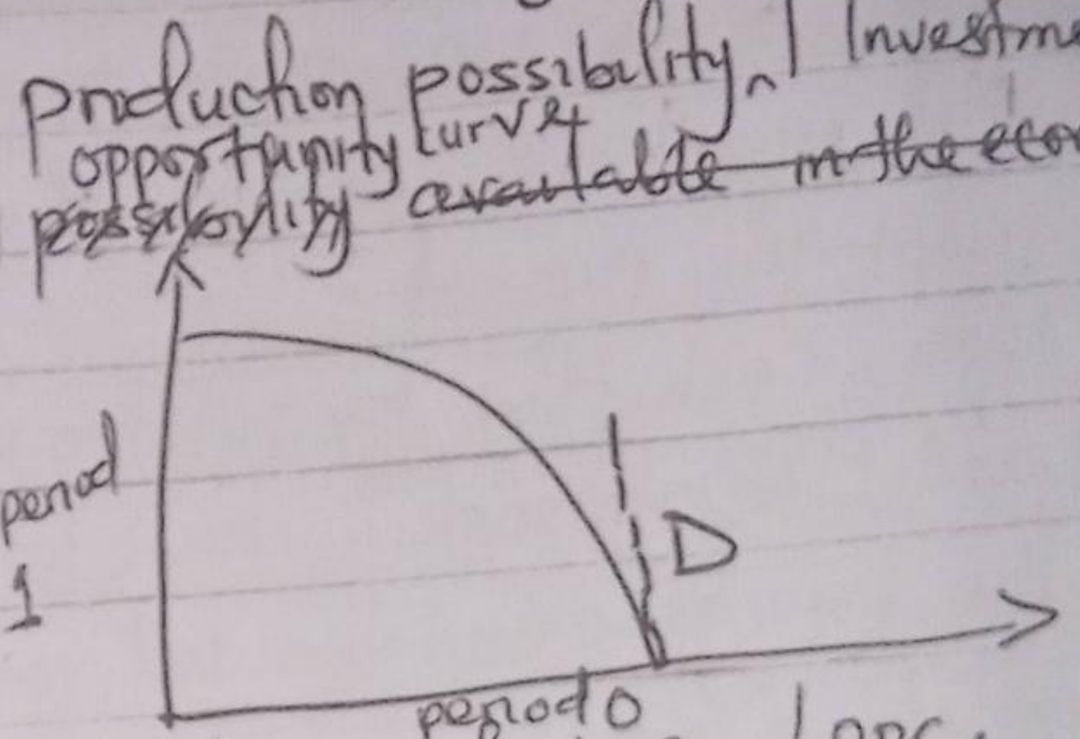
* The higher the indifference, the higher the satisfaction.

* To find the slope, draw a line tangent to the curve. The slope of the curve is the MRS

MRS also called the subjective rate of time preference of individuals

$$MRS = \frac{\partial C_1}{\partial C_0} = -(1+r)$$

Production possibility opportunity possibility curve / Investment available in d econ available in the econon



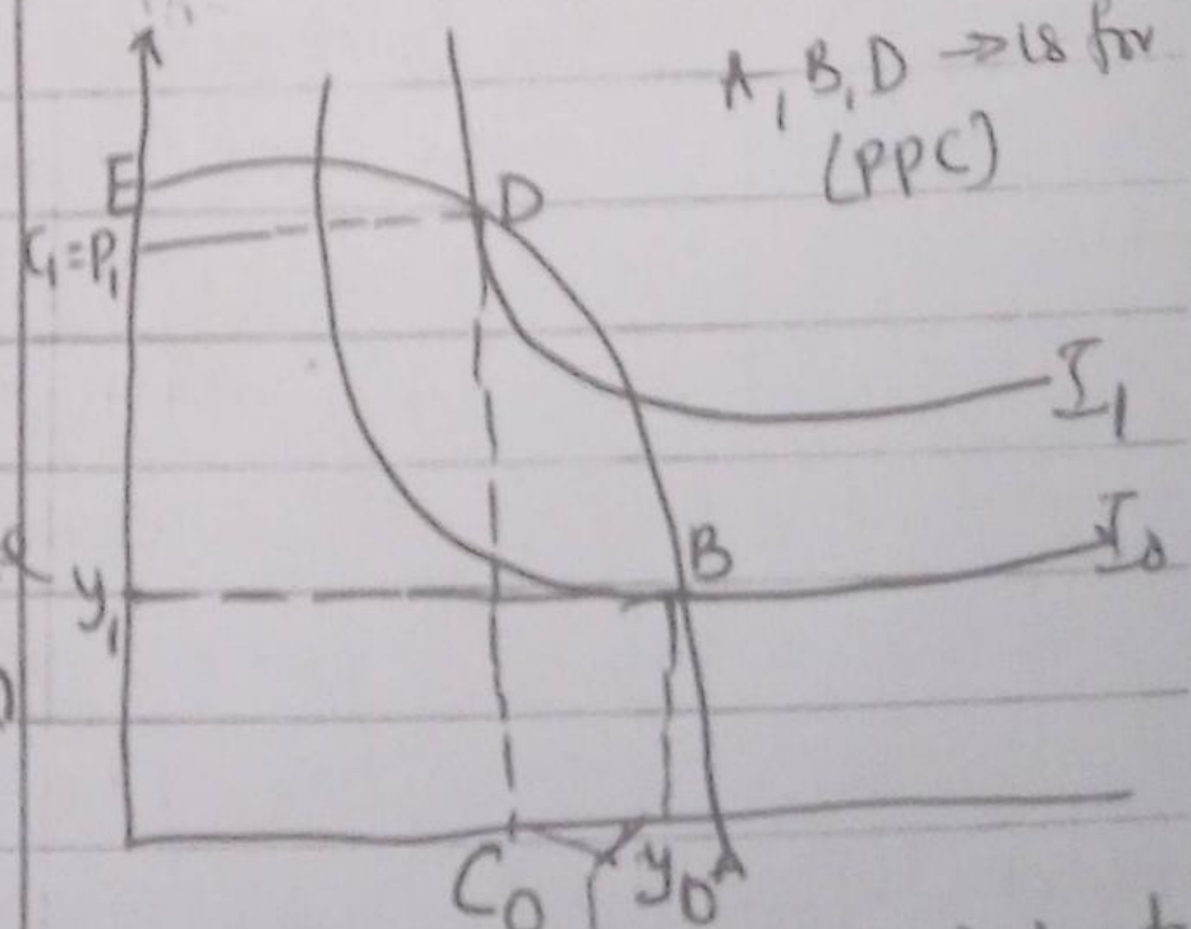
Investment opportunity Curve / PPC.

To find the slope, draw line at a point at the sleepy point "Point D"

The slope is called Marginal Rate of transformation" for PPC curve.

Combination of Indifference Curve and PPC Curve.

To make optimal decision.



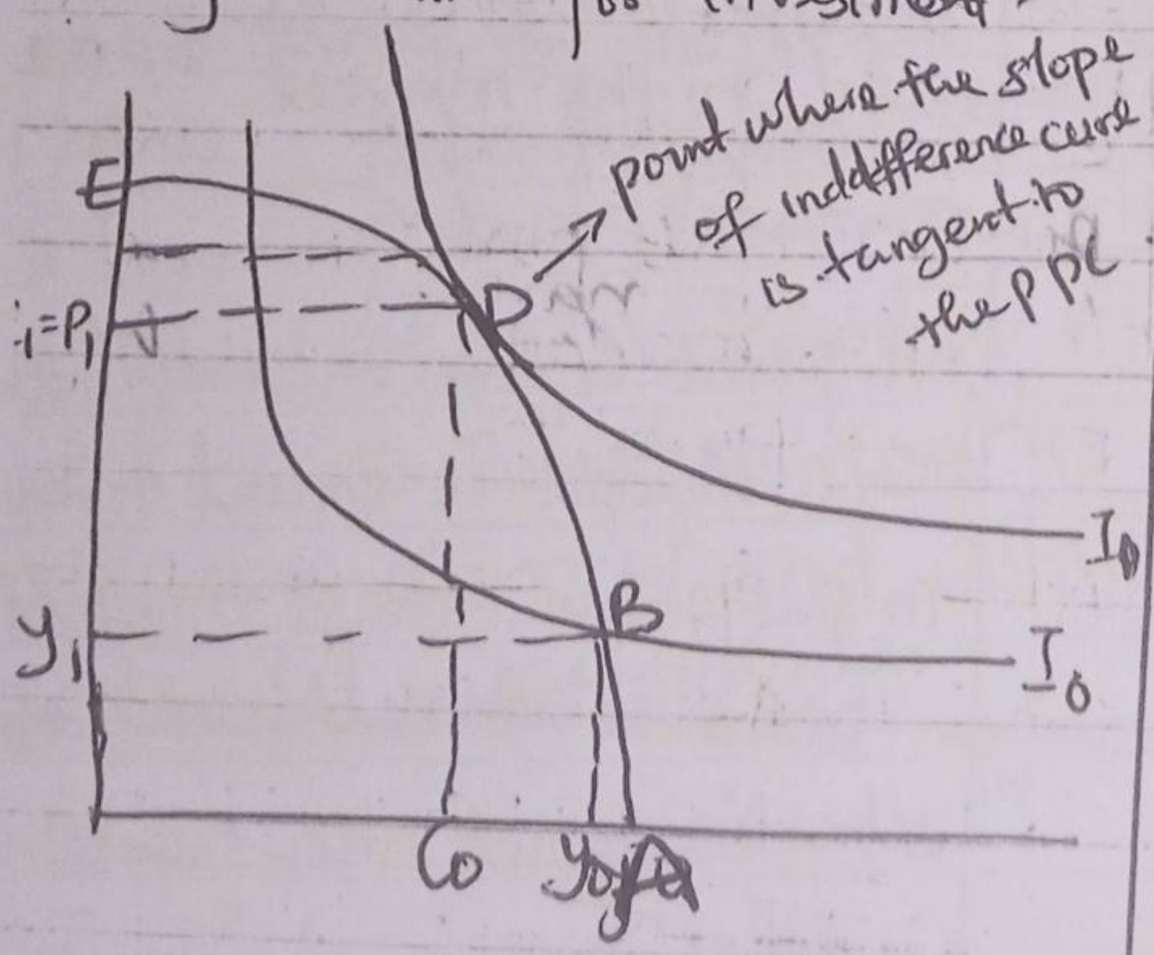
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How to arrive at individual

At point B, individual needs to make decision with his/her initial endowment at point B, he will invest to point D.

At point D, the PPC is tanget to the indifference curve at I_0, Point D is the optimal consuption decision point where MRS = MRT. which is also subjective rate of time preference of an individual

Note! The gap btw Co to Yo is the money remain for investment.



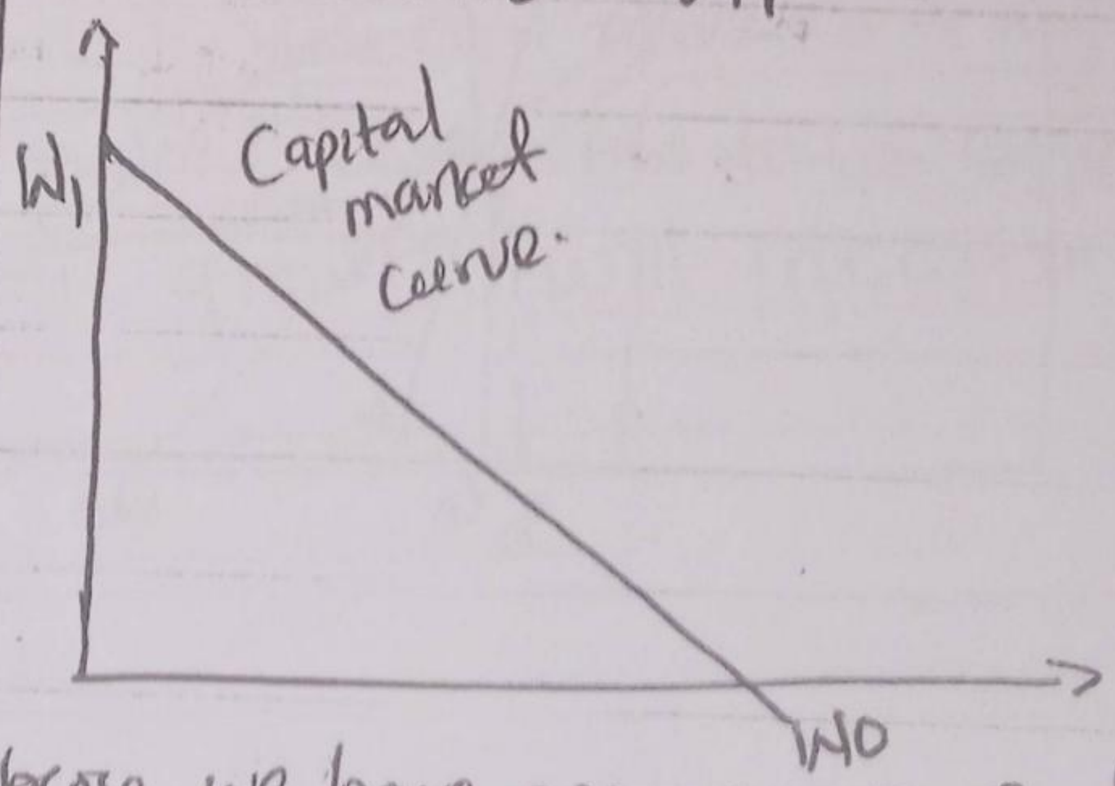
All these are world without capital market.

World with capital Market

We consider the surplus seactors and the deficit seactor in the capital market. Individuals can borrow or lend money out as the capital market serves as an intermediator btw the deficit and surplus seactor. In the capital market, there are changes we will consider or assume the market is a perfect capital market. Under here we assume that lending & borrowing rate are the same

eg If I start at W0, then the interest rate (Wr) after one year it bcos (W1)

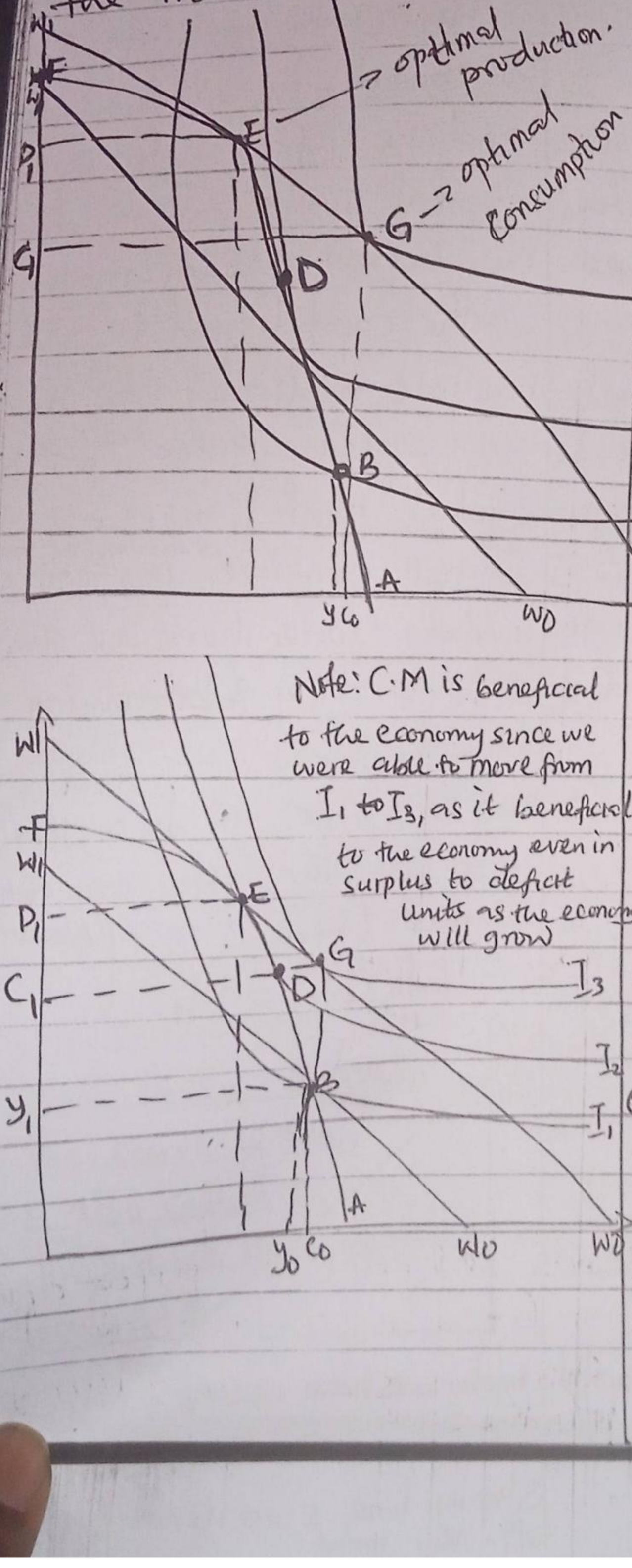
W0 (Wr) = W1 [Assume



bcose we have assume a perfect capital market, we will have a downward sloping line, since lending & borrowing rate the same

Fisher

the slope of the C-market is the "market rate of Interest"



Note: C-M is beneficial to the economy since we were able to move from I_1 to I_3, as it beneficial to the economy even in surplus to deficit units as the economy will grow

Discussion

Point B → Initial endowment Individual has 3 options, eith Consume everything, or Invest in curve I_1, or capital market or Invest in the PPC.

I_3. Point D; this where I_2 is Tangent to PPC where M and he can still continue w Investment by borrowing r from the C-Market to ea higher return to point.

point E: is the maximu point of his Investment. Th the optimal production de where the return on last invest equal to market rate of r Return so at point E, he lend to the market equal to Market until reaches point G.

At point G, the rate of Int of C-Market is equal or tan to subjective rate of time pref individual.

~~Deficit market rate of~~

~~at this point the capital~~

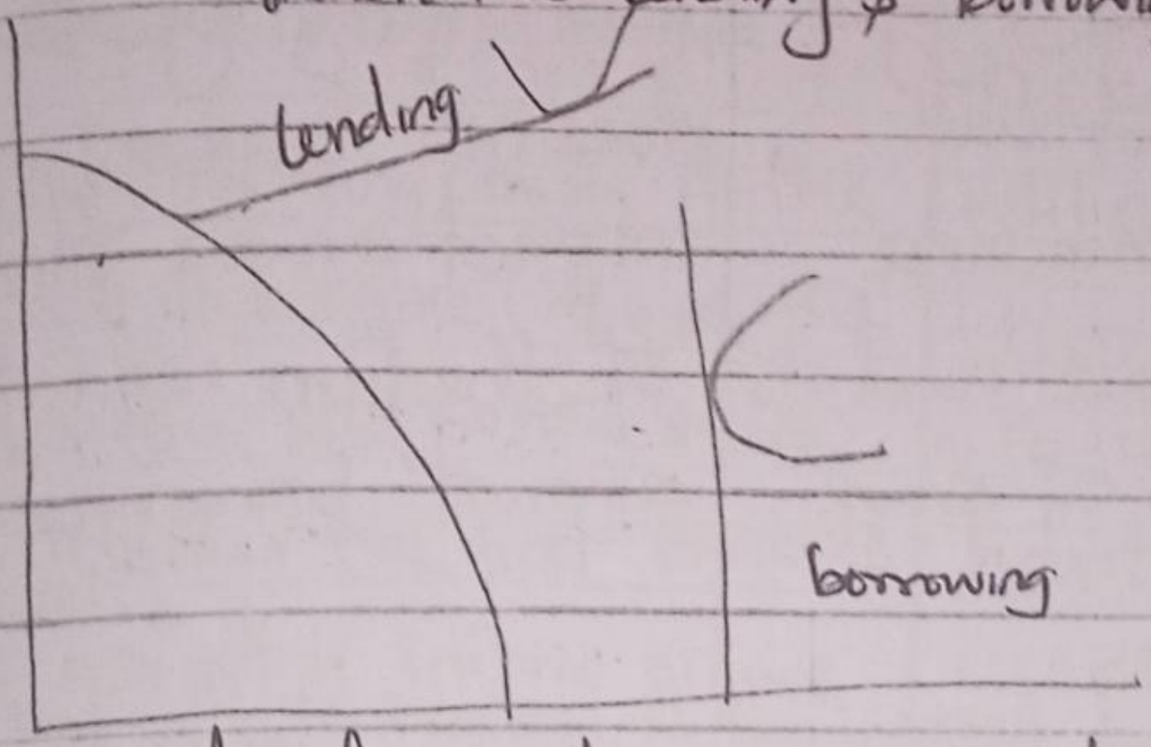
~~is Note~~: Thas analysis is fishers separation the

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Fishers seperation theory.

* Given perfect and complete capital market production decision is govern solely on the basis of objective market criteria without reference to individual subjective rate of time preference which enters into his consumption decision.

When the lending & borrowing rate differs.



Note: Here production decision is at point E, while optimal consumption is at point G.

Implication of this perfect market assumption

1) Lending and borrowing is the same.
2) When there is capital market, you can move to a higher level of consumption and its makes resources to be effiaer ntly transferred and economic growth will increase due to capital market.

27/10/22. DR Amah.

Topic: CAPITAL STRUCTURE DECISION

*In finance key decision that are necessary are (i) viable and efficient ideas (ii) How to fund the investment. the decisions are made using IRR, ARR etc. (finance decision) (iii) Dividend decision

Capital structure is a combination of these sources of capital.

Capital structure only deals with long term source of funds. We are interested in the optimal capital structure of a firm or the target of debt and equity that will give us our target and minimize our cost of finance a bove and also maximise the wealth of share -holders.

Long term funds are called permanent sources of capital.

Financial structure of a firm deals with all the forms of capital and funding in a firm such as short term & long term source of capital. (eg accurals, overdraft etc)

Note: Capital structure is diff from finan -cial structure.

*The higher the risk, the higher the expected return

Optimal C. structure has to do with the best combination of capital structure that result in the maximization of share -holder's wealth and minization of cost of finance)

Theories of capital stru

* Net income theory: This is saying if you want to maximi the wealth of shareholder, the the firm needs to borrow sinc the higher the debt, the high the market value of the firm.

* Net income operating theory: This is saying that debt does not have any significant impact on the market value of the firm regardless of th amount borrowed.

Question: Under Net income theor when cost of debt is constant, cost of equity is constant, why does Weighted Average cost of capital fall?

Note: equity is the most expensive source of capital, very expensive with high return.

Ans: Debt is less risky, with a lower return, note that the debt tax shield helps to reduc the cost of debt

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DR Amah.

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Ans! Debt is less risky, with a lower return, note that the debt should helps to redu the cost of debt

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hence WACC is the important variable we consider.

: Note: As Leverage increases, the WACC falls while the value of a firm increases.

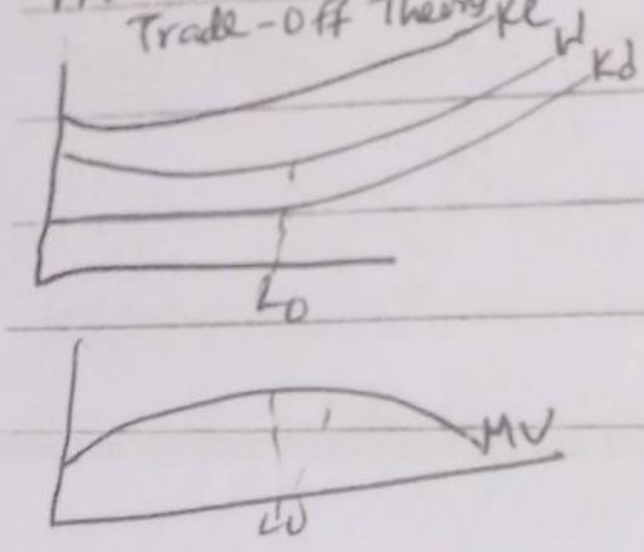
Note: In finance we use market values of book values.

[Revert Capital ~~Market~~ Structure]. Yr 3.

Note: for Net operating coming, If the WACC are the same, the ~~TA~~ the market value will be the same. This is under the first example in the material is for Net Operating Income approach WACC and Kd is constant but Ke is rising). *WACC is also called over all cost of capital.

Note: Pecking Order theory first look at internal sources of capital before getting out (external.

Note: for market value of a levered firm. as WACC, increase market value ↓ / market value decrease vice versa



These are the refinement of the capital structure theory:

I) Traditional or trade-off theory

II) M & M theory.

* In the tradition theory when the Kd and ke starts increasing, this will cause a high risk in the capital structure which will also increase the expectation of the investors. To resolve this the company has to stop it leverage, then use the debt redemption or debt buy back to balance back or when the company is exposed to a more profitable investment opportunity, the company can issue more equity in order to reduce the debt that have sky rocketed the risk of the firm. Bcos the company is at the risk of bankruptcy at the stage after the optimal capital structure on further borrowing is allowed.

At the stage before the optimal C. structure d firm will increase debt in order to increase the MV

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2/11/22. Capital Investment & Tax — Prof. Olowe.

An appraisal that should be done must be after Tax. No Cash loan is related to depreciation. the most disallowed item in the financial statement is depreciation, rather a capital A. will be given

Types of Capital Allowance

*Initial Allowance * ~~Capital~~ Annual Allowance

Initial Allowance: This is given once we purchase an asset. If an asset qualifies an initial allowance, it will be given whether you use it or not.

Annual Allowance: This is given every year. It is given every year until the year of the asset disposal. on the disposal of the ~~asset~~ no allowance is given rather we consider the deed of proceed of the disposal of the asset, it will net up against or compare it with the tax written down value of the asset.

Balancing Charge: If the deed proceed of the disposal of the asset is greater than the tax written down value, then that is the balancing allowance will be given

Capital Allowance can be calculated
* Reducing balance method: At
will be calculated each year.

* Straight line method: same a
is used each year

Illustration: ABC Ltd is consid
a project costing 1 million, the proj
which involves the purchase of cap
is excepted to have a life value o
with a residual value of ₦890
the estimated future cash pro
of the project is as follows
year 1 → 380,000; Yr 2 → 390,
Yr 3 → 400,000; Yr 4 → 405,
capital allowance is available
the capital equipment & project a
follows: Initial allowance 10;
annual allowance on a straigt
line base is 20%, corporate tax i
payable one year in arrers at t
rate of 30%
Advise the Company on whethe
or not to embark on the proj
the company after tax cost
capital is 15%

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Solution.

Step 1: Calculation of capital allowance.

Year	(Cost / NBV at beg or book value)	Allowance.
0	[illegible]	[illegible]
1	Initial (100%)	1000
	Allow (10%)	100
	Annual Allow (20% × 900)	180 — 280
	Written down value →	720
2	Annual A.	(180) — 280
	WDV →	540
3	Annual All	(180) — 180
	WDV	360
4	Deemed proceeds → Residual value	80
	Balance Allowance	280 — 280

Calculation of Tax liability.

Year	1	2	3	4
Cash profit:	380	390	400	405
Less Capital A)	(280)	(180)	(180)	(280)
Taxable profit	100	210	220	125
Tax 3%	(30)	(63)	(66)	(37.5)

Note: The tax of year 1 be paid in year 2 as year one no tax be recorded because the tax is payable one year in arrears

~~recorded / until the end of the year~~ which ~~year~~ first year tax will be calculated ~~in year 2~~

DCF Analysis

Year	Cost	Cash profit)	Tax	Net Cash flow	DF @15%	PV
0	(1000)	—	—	(1000)	1.000	(1000)
1		380	—	380	0.870	330.6
2		390	(30)	360	0.756	272.20
3		400	(63)	337	0.658	221.746
4	80	405	(66)	419	0.572	239.668
5			(37.5)	(37.5)	0.497	(18.638)
						45.16

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Topic: Lease or buy decision. Prof Olaoye. 31/11/22.

Lease is alternative source of funding that usually deal with rent.

Source of fin: * cash * borrowing * leasing
Lease: capital equipment equipment

A Lease is a consideration of rental usually for assets needed for a project. It is an alternative source of funding a project rather than outright buy. It is a "FINANCING DECISION"

It is one means of financing a capital purchase.

Types of Lease.

Note: for every fund you borrow, you pay a fixed interest charges. Ie when you borrow loan to finance your project.

*

1) Operating Lease: This is a short term lease that does not cover most of the useful economic life of the asset eg 3 months after which can be given out again to another person.

Parties in a Lease financing
Lessor: The person that lease out the equipment
Lessee: The debtor or the person taking up the lease contract

Under Operating lease, the "Lessor" bears all responsibility

Finance lease: This is a medium term finance that covers most of the useful economic life of the asset

Periods under F/lease.

* primary period: The rental payable by the lessee covers the cost of the asset and the financial charge. The lessee will be responsible for all the obligation on the asset eg maintenance, insurance.

Secondary period: The lessor can either lease at a less cost, or sell to the lessee or decide to dash it out to the lessee.

Note! At the primary period the lessor has already recov- ered all the cost of the asset

Note! The major people struggling are the lessee as the bank are usually major lessor.

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Note! The major deciding factor before a finance lease can be called a lease is when there is a transfer of ownership. The secondary lease is the main determinant of a lease contract.
According to IFRS for a lease to be qualified as a finance lease, it must have a sale option.
Note: If a lease agreement does not have a sale option, then it is an operating lease.
Note: The secondary lease is the main determinant of a lease contract.
Note: A lessor that refuses to give a sales option claims to be clever due to the capital allowance.
Note! The capital allowance is collected by the lessor.
The major issues in financing a project you must consider different cash flow by carrying out diff Analysis
Another means of financing P.
* Hire purchase: This is sales payable by instalment. But ownership has been transfered owing to the fact that there is no default on the agreement in order to avoide revoke of ownership.

Bcos H purchase is a sales, ownership has be transferred from the day the asset was bought.
How do we evaluate a lease or buy decision!
This is a two way decision i.e Investment decision and financing decision.
Note! If the project is not viable, it can be evaluated, if NPV is (+) you consider the project but for a negative NPV you dont consider. If the project is viable then you give or consider a financing option.
*In evaluating lease or buy decision we are going to evaluate all the financing cash flows from the financing options will be evaluated at the after tax cost of borrow

The above in the next page are the Peculiar cashflow for each financing option. →

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Cash option to purchase:

Initial project cost (project outlay)

Residual value.

Tax saving on capital Allowance

Operating lease option

lease rentals

Tax relief on lease rental.

(no need of initial outlay, no capital allowance, all rent are tax deductible)

Finance Lease option.

Tax saving on capital allowance

Tax relief on Interest element implicit in the lease.

(no initial outlay, there is capital allowance.)

Note! If you borrow to buy, what are the options.

Option to borrow:

In year 0

* project outlay <--> Loan Received

(i.e in year 0, initial outlay is cancelled out by the Loan Borrowed.)

* Tax (saving on capital allowance.
* Residual value.
* Loan Repayment
* Tax Relief on loan Interest payment

Note! When we use Hire purchase, is similar to option to borrow. Under H.P, no need for project outlay that will cancelled by loan received.

Hire purchase option:

Residual value.

Capital allowance

HP. payment.

Tax relief on interest element of hire purchase payment

Illustration! ABC Limited is considering a project costing ₦1 million (CONTINUATION of ABC from in the previous topic) xyz bank has offered to lease lease it capital equip to ABC Ltd as an annual rental of ₦30,000 for 4yrs payable in arrears. ABC currently borrowed at the rate of 21% per annum from the bank.

Should ABC Limited Lease or borrow the capital equipment?

solution

Since there is no sales option, this question is an Operating lease question.

Based on calculation in the previous class we got a +NPV (45.16) showing the project is viable.

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Option to buy. tax = 30% → premium question.

After tax cost of borrowing: 21% (1-0.3) = 14.7 → Discount factor.

Year		₦'00.	DF @ 14.7%	PV
0	Initial outlay	(1,000)	1.0000	(1,000)
4	Residual value	80	0.5778	46.22
Tax savings on capital allowing.				63.85
2	30% × 280,000 = 84	84	0.7601	Year 1 is not included for tax bcas of the airdays
3	30% × 180,000 = 54	54	0.6627	35.79 in the amount
4	30% × 180,000 = 54	54	0.5778	31.20
5	30% × 280000 = 84	84	0.5037	42.31
				(780.63)

Option to Lease.

year		₦'000	DF @ 14.7%	PV
1-4	Lease rental	(300)	2.8724	(861.72)
2-5	Tax relief on Rental 30% × 300,000	90	2.5043	225.387 (636.33)

$\left[\frac{1-(1+r)^{-n}}{r}\right]$ where n=4

Note: The best option is the option to lease, since it has a lesser negative NPV

Dif for 2-5 $= \left[\frac{1-(1+0.147)^{-5}}{0.145}\right] - \left[\frac{1-(1+0.147)^{-1}}{0.145}\right]$

NEXT TOPIC: Utility theory.

Utility theory is a theory of choice.

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Note: The ~~best~~ option is the option to lease, since it has a ~~positive~~ lesser negative NPV

$$\text{Diff for 2-5} = \left[\frac{[1-(1+0.147)^{-5}]}{0.145}\right] - \left[\frac{[1-(1+0.147)^{-1}]}{0.145}\right]$$

NEXT TOPIC: Utility theory.

Utility theory is a theory of choice, choice among risky asset

to have better understanding of U. Theory, you must consider the ~~U~~underlying of assumptions of U-Theory

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Cardinal point of U-Theory / Assumptions of Utility theory or Axioms

1. Comparability : for an entire set of X alternative, an individual can compare item X to Y or Y to Z. <i.e there are two things you are trying to compare from. 2 things or options are available.

2) Consistency : If outcome X is preferred / indifferent to Y and outcome Y is preferred / indifferent to Z, then X must be preferred / indifferent to Z.
   If $X \succ Y$, and $Y \succ Z$ then $X \succ Z$ <preferred).
   or If $X \sim Y$ and $Y \sim Z$ then $X \sim Z$. <indifferent btw X and Z

3) Strong Independent : Suppose we set up a gamble such that it has probability α of receiving outcome x and $(1-\alpha)$ of receiving outcome Y. :: $G(X, Z; \alpha]$
   If x is indifferent to Y, then gamble set up btw X with prob α and Y with prob $(1-\alpha)$ (check)

Then: $G(X, Z; \alpha)$ in $G(Y, Z; \alpha)$ this is strong independent

4) Measurability : If X is preffered more than Y, and Y is preferred more than Z i.e there is a unique α such that Y indefferent btw Gamble $X, Z; \alpha$.
   If $X \succ Y \succ Z$, there is a unique α such that.
   $Y \sim G(X, Z; \alpha)$.

5) Ranking : this axiom follows from measurability. If it can be est that X is preferred more than
   If $X \succ Y \succ Z$ and $X \succ U \succ Z$
   then $Y \sim G(X, Z; \alpha_1)$; $U \sim G(X, Z; \alpha_2)$.
   If $\alpha_1 > \alpha_2$, then $Y \succ U$

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$\alpha_1 \geq \alpha_2$,

$\alpha_1 < \alpha_2$, then $U > y$

you must add to this assumption, the fact that human being are greedy bcos Every individual wants to maximize their expected Utility of wealth.

Note $G(x, z; \alpha)$.

prob $x = \alpha$; then Prob $z = (1-\alpha)$

$E(w) = G(x)\alpha + G(z)(1-\alpha)$

↓ expected wealth.

To get the expected Utility of wealth

$E[U(w)] = U(x)(\alpha) + U(z)(1-\alpha)$

$E[U(w)] = \sum_{i=1}^{n} U(x_i)\ Prob(x_i)$

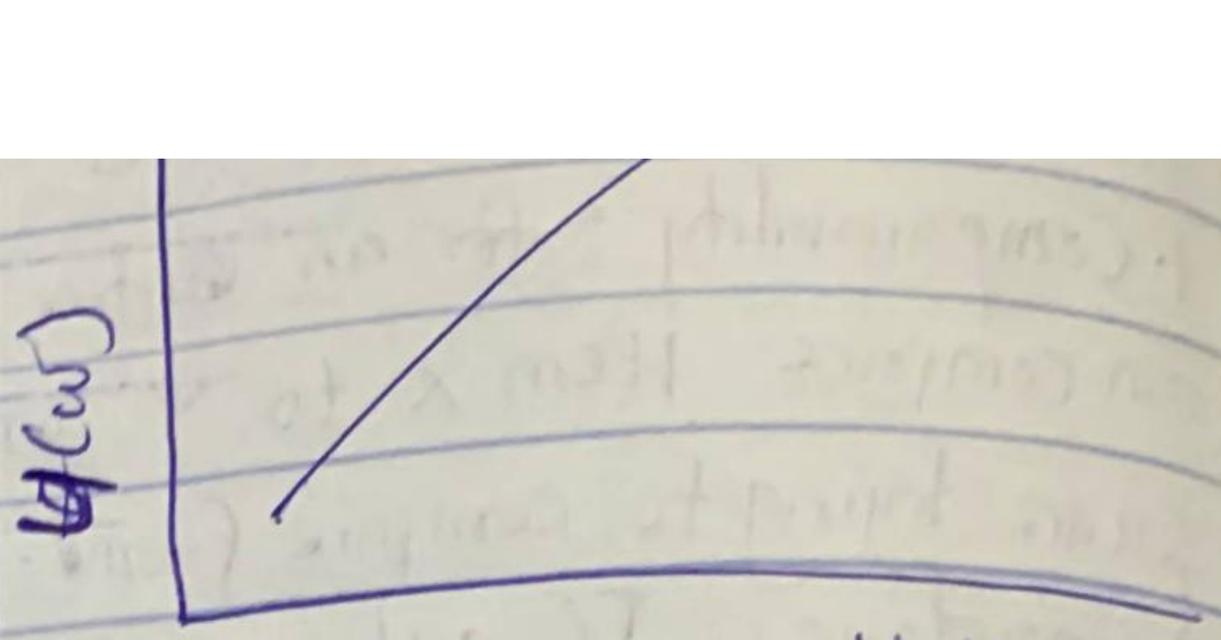
Utility Function.

This U. function can take up any case

Utility function is stated as the function of wealth. And we have various Wealth Level

W =	10	20	30	50	60
U(w)	1.5	2.0	2.5	3.5	4.0

Note: Once I plot my utility function I can determine my risk attitude



It can be any of the following:

i) Linear function : $U(w) = a + bw$

ii) quadratic fun : $U(w) = a + bw + cw^2$

iii) cubic : $U(w)\ a + bw + cw^2 + dw^3$

iv) logathrim F: $U(3) = \ln w$

Attitude to Risk

Risk lover : A person that prefer taking a gamble

Risk averse : A person that perfers taking actual value of the averg -age outcome

Risk neutral : A person that prefers to take risk or not.

Illustration: Doyin is considering taking a gamble that increase in wealth by 10,000, Pro (60%) and another gamable that increase her wealth by ₦500 with a prob (40%

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Olu has a current wealth of 10,000 and a logarithm utility function What is his attitude to risk.

solution

Since the current wealth is 10,000 then once he gambles, there will be an increase in wealth.

New wealth = 10,000 + 10,000 = 20,000

50% of 10000 = 10,500

$E(w) = 20,000(0.6) + 10,500(0.4$

$= 16,200$

$U[E(w)] = 9.69$ util. → Ln(16,200)

$E(U(w)) = U(20,000)(0.6) + U(10,500)(0.4)$.

Ln(20,000) Ln(10,500)

$= 9.9(0.6) + 9.26(0.4)$.

$= 9.644$ utils.

If $U[E(w)] > E(U(w)]$ → risk averse.

If $U[E(w)] = E(U(w)]$ ⇒ risk neutral

If $U[E(w)] < E[U(w)]$ ⇒ Risk lovers

where $U[E(w)]$ = Utility of Expected wealth

where $E(U(w)]$ = Expected Utility of wealth.

Conclusion: Doyin is risk averse because utility of expected wealth is greater than expected utility of wealth.

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